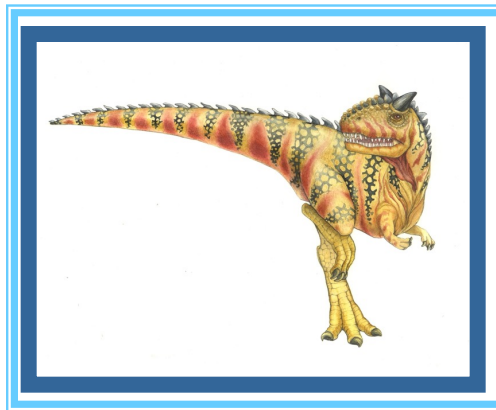


Chapter 2: Operating-System Structures





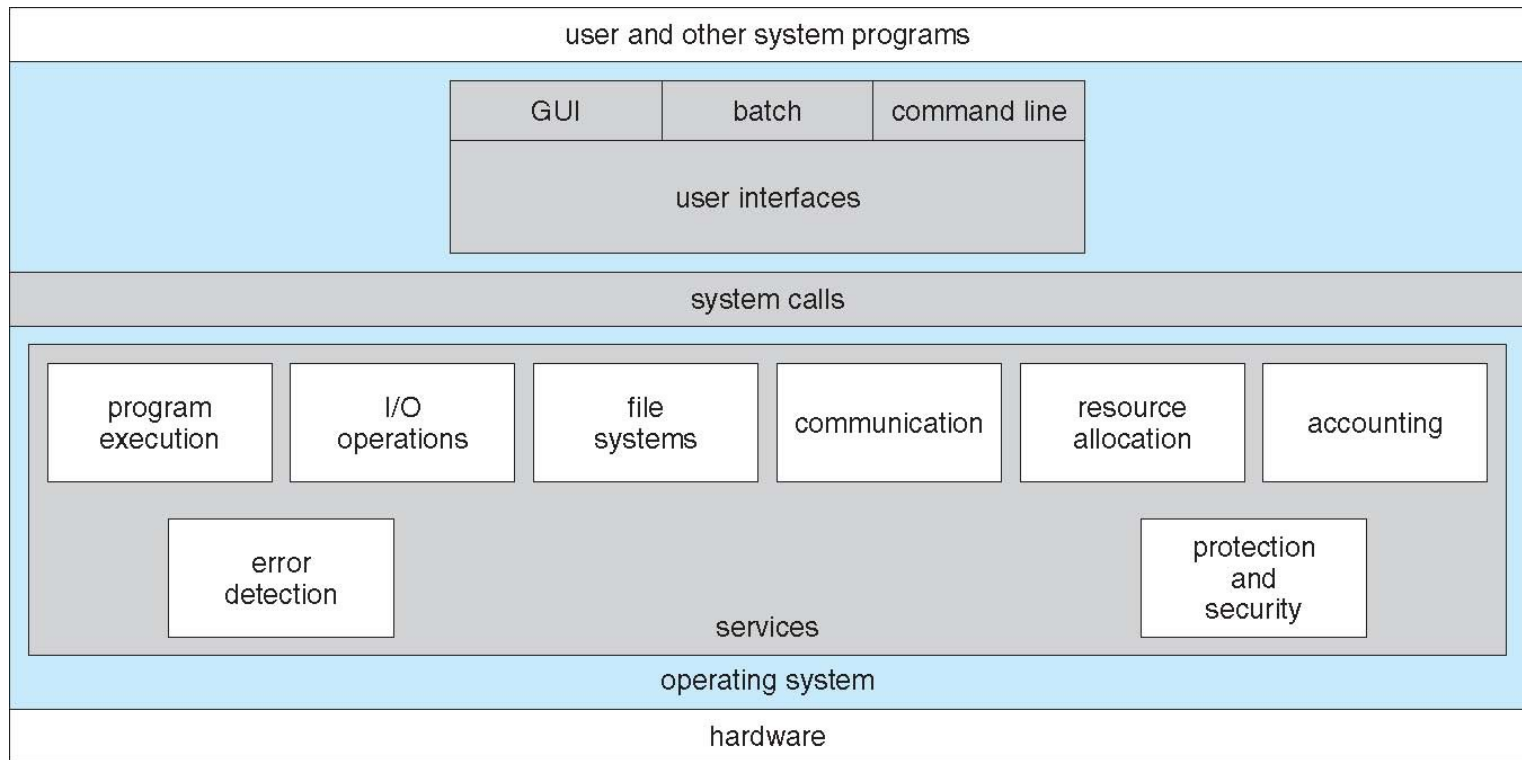
Operating System Services

- One set of operating-system services provides functions that are helpful to the user:
 - User interface - Almost all operating systems have a user interface (UI)
 - ▶ Varies between **Command-Line (CLI)**, **Graphics User Interface (GUI)**, **Batch**
 - Program execution - The system must be able to load a program into memory and to run that program, end execution, either normally or abnormally (indicating error)
 - I/O operations - A running program may require I/O, which may involve a file or an I/O device
 - File-system manipulation - The file system is of particular interest. Obviously, programs need to read and write files and directories, create and delete them, search them, list file Information, permission management.





A View of Operating System Services





Operating System Services (Cont)

- One set of operating-system services provides functions that are helpful to the user (Cont):
 - Communications – Processes may exchange information, on the same computer or between computers over a network
 - ▶ Communications may be via shared memory or through message passing (packets moved by the OS)
 - Error detection – OS needs to be constantly aware of possible errors
 - ▶ May occur in the CPU and memory hardware, in I/O devices, in user program
 - ▶ For each type of error, OS should take the appropriate action to ensure correct and consistent computing
 - ▶ Debugging facilities can greatly enhance the user's and programmer's abilities to efficiently use the system





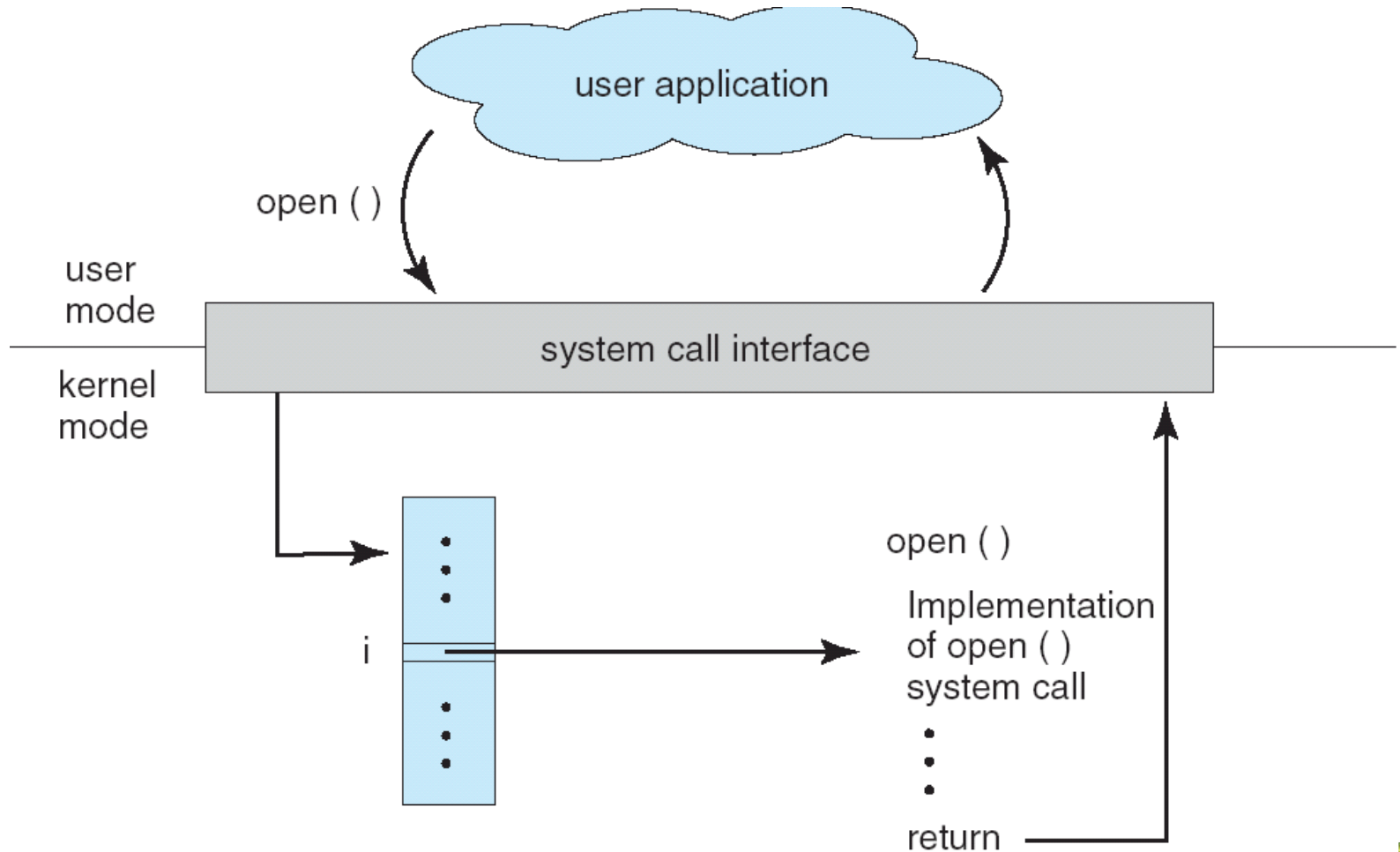
Operating System Services (Cont)

- Another set of OS functions exists for ensuring the efficient operation of the system itself via resource sharing
 - **Resource allocation** - When multiple users or multiple jobs running concurrently, resources must be allocated to each of them
 - ▶ Many types of resources - Some (such as CPU cycles, main memory, and file storage) may have special allocation code, others (such as I/O devices) may have general request and release code
 - **Accounting** - To keep track of which users use how much and what kinds of computer resources
 - **Protection and security** - The owners of information stored in a multiuser or networked computer system may want to control use of that information, concurrent processes should not interfere with each other
 - ▶ **Protection** involves ensuring that all access to system resources is controlled
 - ▶ **Security** of the system from outsiders requires user authentication, extends to defending external I/O devices from invalid access attempts
 - ▶ If a system is to be protected and secure, precautions must be instituted throughout it. A chain is only as strong as its weakest link.





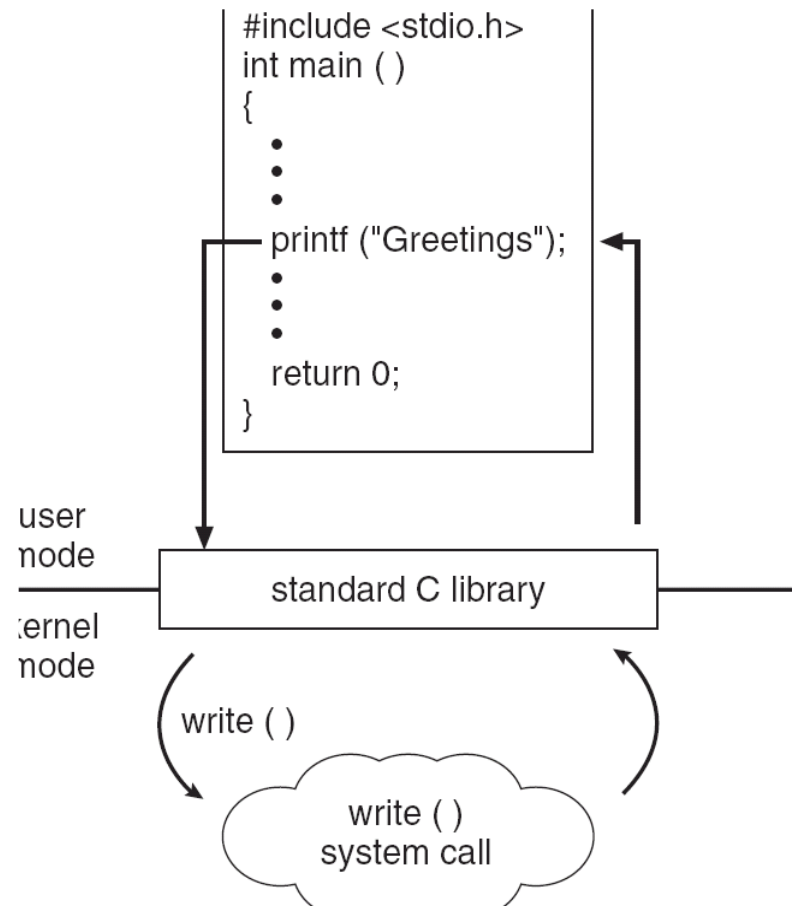
API – System Call – OS Relationship





Standard C Library Example

- C program invoking printf() library call, which calls write() system call





System Call Parameter Passing

- Often, more information is required than simply identity of desired system call
 - Exact type and amount of information vary according to OS and call
- Three general methods used to pass parameters to the OS
 - Simplest: pass the parameters in *registers*
 - ▶ In some cases, may be more parameters than registers
 - Parameters stored in a *block*, or table, in memory, and address of block passed as a parameter in a register
 - ▶ This approach taken by Linux and Solaris
 - Parameters placed, or *pushed*, onto the *stack* by the program and *popped* off the stack by the operating system
 - Block and stack methods do not limit the number or length of parameters being passed





Parameter Passing via Table

